

So you're getting a data centre! Here's what to know | CCPA

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Share Data centre construction is booming. Canadians in towns and cities across the country are waking up to the prospect of a data centre as their new neighbour—albeit a neighbour who is a fair bit more energy and water hungry than the average Canadian residence. In the United States, where data centre investment is in full swing, governments across the country—fearing they will miss out on the data centre investment boom—are trying to lure developers to their areas with a mix of tax incentives and subsidies. At the same time, some communities are expressing growing concern over the impact of data centres and fighting against their own governments to prevent them from being constructed. As Canadians begin to experience their own data centre boom—driven by the computing needs of Artificial Intelligence (AI) and the need for data sovereignty—we need to understand what we are getting into and what questions to ask when we invite data centres to be our neighbours. Will my power bill go up? Whether a data centre will increase your electricity bill is one of the primary concerns people raise when companies or governments propose building a data centre in their community. There is no doubt that data centres are voracious energy consumers, but their individual energy needs depend on the size of the centre, the amount of servers, the cooling system and the kind of computing the centre performs (AI training, for example, is a lot more energy intensive than cloud computing). Small data centres, typically spanning 5,000-20,000 square feet with 500-2,000 servers, consume between one and five megawatts (MW) of power. These facilities often serve single organizations or provide co-location services for smaller businesses. A medium-sized facility may consume up to 20 MW. However, a single large hyperscale data centre can consume up to 100 or more MW of power continuously, equivalent to powering 15,000-75,000 homes. And these hyperscale centres are only getting bigger. The proposed data centre campus in Olds, Alberta would consume one gigawatt (GW) of power, enough to power 750,000 homes. Providing the needed energy for these centres can dramatically increase electricity demand and require utilities to pass along increased infrastructure upgrade and new generation costs onto customers. Residents close to “data center alley” in Northern Virginia have seen their power bills double as data centres drive up demand for electricity. Realizing the costs these centres are imposing on residents, certain state governments are now requiring that data centre developers foot the majority of the bill for grid upgrades and new generation costs. This is an example provincial governments might want to emulate to ensure that new data centre construction does not exacerbate the affordability crisis already affecting many Canadians. Question to ask: - Who will pay for increased electricity costs brought on by increased demand or the cost of grid upgrades and new electricity generation? What about water? Data centres also

need water for the cooling systems that are required to control the intense heat created by high-powered processors. Once again, the amount of water required will depend on the size of the data centre, the cooling system it employs and the climatic conditions it operates in. (Due to its cold climate, Canada is considered an ideal location for data centres). Water consumption estimates are further complicated by the fact that many data centres in the U.S. do not have to disclose how much water they use. A mid-sized data centre consumes around 300,000 gallons (1.1 million litres) of water a day, or about as much as 1,000 households. However, relative to other commercial consumers of water, data centres do not appear to be profligate consumers. For example, a single golf course can consume between 100,000 to two million gallons per day. The consensus appears to be that regions that are already under significant water stress, like Alberta and Saskatchewan, will have to determine what industries will have priority when it comes to the allocation of scarce water resources—and whether data centres deserve priority over other consumers. Questions to ask: - Will this data centre be required to disclose its daily water consumption to the public? - What policies are in place to conserve and/or re-use water? Are they noisy? One of the more common complaints of residents that live close to data centres is the constant noise from fans, turbines, generators, etc. While centres usually operate under noise law thresholds, the constant low frequency drone of these facilities have reportedly disrupted sleep and caused health problems for some residents that live nearby. The problem is that most noise ordinances weren't written with 24-hour data centre operations in mind. Decibel thresholds that might be reasonable during the day, may need to be reconsidered to address the constant noise of these centres in the dead of night. Questions to ask: - What kinds of noise suppression systems will this data centre have in place? - Will there be a distance requirement between data centres and residences, hospitals or schools? - Are current noise by-laws adequate to address the unique noise pollution of a 24-hour data centre? What about electronic waste? In addition to water and electricity consumption, data centres also generate hazardous waste in the form of electronic waste or “e-waste.” Servers, batteries, cooling systems and other hardware contain toxic materials and refrigerants that can leak harmful chemicals into soil and water when improperly handled or disposed of. E-waste is considered hazardous because it contains toxic substances that have been linked to serious health problems such as cardiovascular disease, respiratory damage, and cancer. It is also the fastest growing type of solid waste globally and is often overlooked in discussions about data centre development. Studies estimate that the computational materials used to develop generative AI could generate up to five million tonnes of e-waste by 2030. Questions to ask: - Where will this hazardous waste go? - Who is responsible for safe disposal? - And will local governments be left to manage the long-term environmental risks? Will it create jobs? The promise of data centre developers is that these facilities will create jobs and spur investment in the local community. So are data centres also job creators? During the construction phase, which can last from 12 to 18 months, approximately 1,500 workers are on site in some capacity. However, once operational, data centres simply don't employ many workers. Labour expenditures once operational only represent about 15 per cent of the total output of a data centre. The Government of Virginia estimates that a typical size data centre in that state employs 50 workers, with half of those usually contract workers employed in security, maintenance and janitorial work. Indirect job creation is also limited due to the nature of the data centre. Temporary construction jobs can have spill-over effects for local building industries and businesses that service workers. But as with direct job creation, once the centre is built, spin-off job creation seems small. Data centre equipment is highly specialized, so not something that can be easily sourced from local or even regional businesses. Nathan Jensen, a professor at the University of Texas at Austin, who researches economic development strategies, argues that “unlike many industries, like automobile manufacturing, there isn't a clear case for suppliers locating near data centres,” he said. “They don't really need much of an ecosystem around the facility.” Data centre developers often over-promise or adjust job estimates in order to placate local opposition, so job estimates should be taken with a large grain of salt. Questions to ask: - How many permanent, full-time jobs will the data centre create? - How many of these jobs will be high-skill and high-paid? - How many of these jobs can be sourced

from the local community? Will they provide tax revenue? Another frequent promise of data centres is the tax revenue they will bring to local communities. Certainly data centres can provide significant tax revenues as long as governments do not sacrifice those taxes in a bid to attract the data centre in the first place. Governments across the U.S. have forgone billions in state and local taxes in a bid to lure data centre investment. These exemptions rarely pay for themselves. An audit of the economic impact of data centres in Virginia estimates the state generated 48 cents in new state revenue for every dollar it did not collect in sales tax on data centres between fiscal years 2014 and 2023. Moreover, providing tax exemptions to a facility that may very well increase the cost of living for its surrounding residents would seem to be a poor use of public policy. There is also the question of what will be taxed and by whom. The most expensive part of a data centre is often the equipment within it, up to 70 per cent of the average capital expenditure on a data centre is information technology. Alberta—which doesn’t have a sales tax—has opted for a two per cent levy on the value of computer hardware within a data centre greater than 75 MW. However, this means that for municipalities in Alberta, only land and buildings associated with data centres will be subject to assessment and taxation, reducing the potential tax revenue for municipalities, as this hardware “carries the highest assessable value.” Questions about how tax revenues will be distributed between different levels of government are equally important. Local governments may be left on the hook for investments to service facilities without the ability to recoup these costs. Questions to ask: - What kinds of incentives (tax breaks, land cost, power subsidies, regulatory relief) have we offered to developers to construct this data centre here? - Will these incentives cover the cost of building and servicing the facility for however many years it is in operation? What regulations do we need? We need more than just promises when data centres come to town, we need real guarantees that the benefits of these new neighbors will be maximized and the costs to our communities will be minimized. Here are some ideas that other communities have proposed. - Utilities must provide rate protections or discounts to communities hosting new data centres to prevent residents from absorbing higher electricity and water costs driven by data centre demand. Data centres should pay a portion of the costs for grid upgrades or new generation capacity. - A Mandatory Community Benefit Agreement before permits are issued would allow direct, meaningful representation from affected communities. Agreements should secure concrete local benefits, not just consultation (job guarantees, skills training, local suppliers). - Update noise by-laws to address the unique noise pollution of data centres. Require data centres to use sound baffling or other noise suppressants on equipment. Distance requirements from homes, schools, long-term care facilities and hospitals should be considered. - Prohibit tax breaks, subsidies, or other public incentives for data centres powered by fossil fuels. Public funds should not support new emissions. - Require public reporting of energy use, water consumption, and emissions to ensure accountability and allow assessment of cumulative impacts. - Require binding offset plans addressing land use, water stress, and emissions (could look like green infrastructure, reforestation, water conservation, etc). Offsets must be local, verifiable, and tied to an ongoing monitoring framework. Data centres are coming, and their construction is likely to increase at a rapid pace in the coming years. It is increasingly important that governments at all levels attempt to maximize the social benefits from their construction and operation while minimizing their well-documented negatives. To do that, make sure that your elected representatives have well-thought out answers to all of the questions we asked today.